MONO SINGLE-ENDED CLASS-A | 40 V SINGLE SUPPLY / CAP-COUPLED | 8 ohm / 12-15 W target

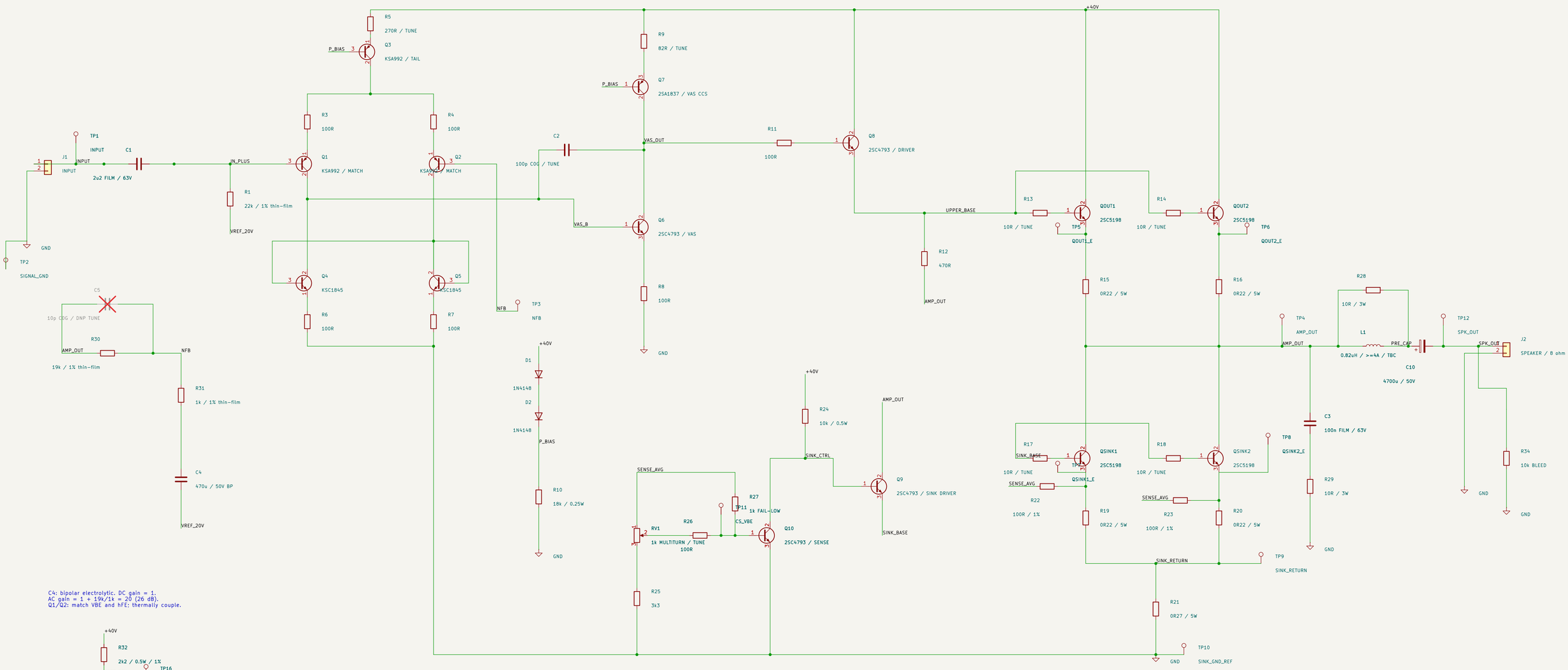
INPUT / DIFFERENTIAL PAIR

CURRENT MIRROR / VAS

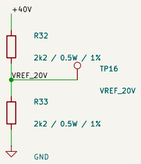
UPPER SIGNAL DRIVER

NPN CLASS-A POWER STAGE

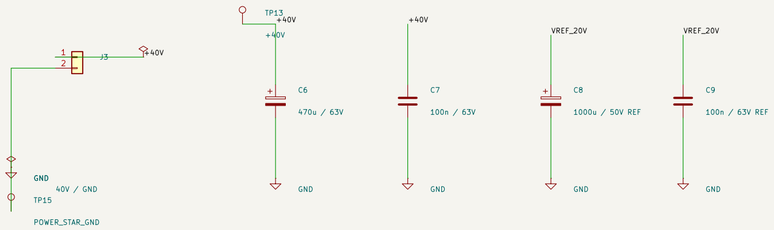
OUTPUT / SPEAKER



C4: bipolar electrolytic. DC gain = 1.
AC gain = 1 + 19k/1k = 20 (26 dB).
Q1/Q2: match VBE and hFE; thermally couple.



EXTERNAL 40 V SUPPLY / LOCAL DECOUPLING



GROUND: one GND net. Separate input/feedback returns from speaker/Zobel and rail-cap returns.
Join at J3 GND/local capacitor star. VREF is a quiet signal reference, not a power return.

40 V single supply. AMP_OUT rests at ~20 V DC; SPK_OUT has DC blocked by C10.
C10 positive faces amplifier; 4700u / 50V gives ~4.23 Hz into 8 ohm.
VREF = filtered divider midpoint (~20 V): input/feedback only, never speaker return.
C1 must remain fitted for a ground-referenced source; DC bypass R2 removed.
C8/C9 filter VREF; C6/C7 bypass the 40 V supply. R10 changed to preserve bias.
Startup/shutdown thump and reference settling require bench verification.

ALL FOOTPRINTS UNASSIGNED: exact owned package/lead-form verification required before PCB.
Pins: 2SC5198 / 2SC4793 / 2SA1837 = 1 B, 2 C, 3 E; KSC1845 / KSA992 = 1 E, 2 C, 3 B.

CURRENT SINK SETUP / THERMAL

RV1 wiper toward SENSE_AVG = minimum current. Set unloaded and current limited.
Set each R19/R20 to 220 mV (1.00 A each); R21 ~540 mV (2.00 A total).
I = VBE(Q10) / [alpha x (0.27 + 0.22/2)]; alpha ~0.79 to 1.
Nominal range ~1.6-2.2 A; junction temperature/device spread: TBC.
R27 pulls Q10 base toward full sense on open wiper: current decreases.
Thermally couple Q10 to lower-bank heatsink; hotter Q10 reduces current.
QOUT1/2 and QSINK1/2 need substantial external heatsinking and
electrical isolation: Toshiba 2SC5198 collector tabs are electrically live.
Match parallel devices; each has its own non-inductive 0R22 ballast and 10R stopper.

FIRST-PASS LIMITS / BENCH TBC

Output-stage idle supply power ~80 W: ~19-20 W per power transistor.
12 W: 13.86 Vpk / 1.73 Apk. 15 W: 15.49 Vpk / 1.94 Apk.
2 A sink gives ideal resistive Class-A current ceiling ~16 W / 8 ohm.
Upper bank peaks ~3.94 A: budget beta=35 => ~113 mA base drive.
Q8/Q9 require thermal margin: Q6/Q7 ~8 mA, ~0.16 W each at idle.
C2 100p, C5 DNP stoppers and L1: bench stability/step response TUNE.
AMP_OUT ~20 V DC; speaker output AC coupled. Test capacitor charging,
rail sag, sink transient response and hot DC SOA before PCB layout.
No simulation, bench qualification, PSU design or PCB work in this revision.

Sheet: /	
File: ClassA.kicad_sch	
Title: Mono single-ended Class-A / 2SC5198	
Size: User	Date: 2026-10-06
KiCad E.D.A. 10.0.5	**Rev: 0.2 SINGLE SUPPLY**
	Id: 1/1